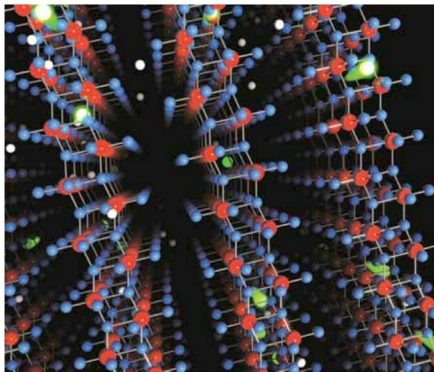




2D material for ultra-high speed electronics (Courtesy: <http://www.pdnet.com/>)

Describing the functioning of wires, Dickey shared, when the wires are sliced or severed, the liquid metal oxidises to form a 'skin' that prevents it from leaking out of its sheath. When the severed edges of the wire are placed back together, the liquid metal reconnects and the sheath re-forms its molecular bonds.

According to scientists, this research will pave the way for creation of more complex circuits and three-dimensional structures.

Bendable batteries

After bendable smartphones and displays, South Korean scientists have now developed the world's first bendable lithium-ion batteries, which may lead to the development of other flexible mobile devices. To create the battery, researchers led by Professor Lee Sang-young of the Ulsan National Institute of Science and Technology used imprintable, fluid-like polymer electrolytes. The team actually spread the fluid-like electrolytes on electrodes and exposed them to ultraviolet rays for 30 seconds.

Informing about this invention, an official at the South Korean Education, Science and Technology ministry said, "Conventional lithium-ion batteries that use liquefied electrolytes had safety problems as the film that separates the electrolytes may melt under heat, in which case the positive and negative elements may come in contact, causing an explosion."

The official added, "Because the new battery uses flexible but solid materials—and not liquids—it can be expected to show a much higher level of stability than conventional rechargeable batteries."

Microscope to detect objects at nano scale

Researchers at the UCLA's Henry Samueli School of Engineering and Applied Science have created a nano-lens microscope using which objects smaller than 100 nanometres could be detected. With the help of optical microscopy method developed by the researchers, one would be able to directly see even those objects which are over a thousand

times smaller than the width of a human hair.

The new microscope consists of tiny liquid lenses—typically thinner than 200 nanometres—which self-assemble around microscopic objects on a glass substrate. According to the researchers, it could be beneficial for diagnosis of diseases in point-of-care settings or areas where medical resources are limited.

Describing the functioning of the microscope, lead author Aydogan Ozcan said, "A simple light source, such as a light-emitting diode (LED), is used to illuminate the nano-lens object assembly. By utilising a silicon-based sensor array, lens-free holograms of the nanoparticles are detected. The holograms are then rapidly reconstructed with the help of a personal computer to detect single nanoparticles on a glass substrate."

2D material to create ultra-high speed electronics

Paving the way for creation of ultra-high-speed electronic products, researchers at CSIRO and RMIT University, Australia, have created a two-dimensional material from graphene that allows electrons to flow at an ultra-high speed.

Graphene has some physical properties that prevent its use in high-speed electronics. The team led by Dr Serge Zhuiykov was able to overcome this disadvantage with the help of layers of a crystal called molybdenum oxide.

Zhuiykov explained that the new nano-material, which is 11 nm thick, is made up of layers. Within the layers, electrons are able to zip through at a high speed with minimal scattering.

Researchers manipulated the material to convert it into a semiconductor. Then they created nanoscale transistors using molybdenum oxide. As a result, they successfully achieved electron mobility values greater than 1100 cm²/Vs—exceeding the current industry standard for low-dimensional silicon.